

RAG-to-Riches: Mining Financial News Signals for Multi-Class Stock Return Prediction

Rashed Aldulijjan
Kaleb Aklilu
Sristi Bafna
Yingxi Su
Yujie Lin
Johns Hopkins University
May 2026

Abstract

We present **RAG-to-Riches**, an end-to-end pipeline for predicting discrete short-term stock returns by combining financial news signals with classical technical analysis. Starting from raw news articles retrieved via Google BigQuery’s Global News Knowledge Graph (GKG) and scraped at scale across 20 distributed worker nodes, we construct a daily stock-level feature table spanning 10 equities over nine years (2017–2026) with 1,145 raw feature dimensions. News articles are filtered for ticker-relevance, scored with FINBERT for three-class sentiment, encoded with a 1,024-dimensional sentence-transformer (*e5-large-v2*), and further enriched through structured event extraction using a local large language model. These signals are merged with 86 technical indicators derived from OHLCV price data to form a 20,355 stock-day dataset. We benchmark four model families, a regression neural network, a binary classifier, residual deep neural networks, and gradient-boosted trees, on a temporally ordered five-class return-bucket prediction task. Gradient-boosted trees (CATBOOST with PCA-compressed embeddings) outperform all neural baselines, achieving **62.5%** test accuracy and a **60.4%** macro-F1 score, compared to 24.8% for the best deep neural variant. Our results underscore the continued strength of gradient boosting for mixed-modality tabular finance data and highlight the complementary nature of news-derived semantics and price-derived technical features.

1 Introduction

The efficient-market hypothesis posits that asset prices reflect all publicly available information [5]. In practice, however, information dissemination is uneven: financial news is published continuously, markets react with latency, and investors differ in their ability to process large volumes of text. This creates a short-lived but potentially exploitable gap between news content and price adjustment that machine learning systems can target [13].

Predicting the direction or magnitude of near-term stock returns from news has a long history in computational finance [2, 12]. Recent advances in pre-trained language models [17, 7] have made high-quality financial sentiment analyses accessible, while transformer-based dense retrievers enable rich semantic encoding of article content at scale. At the same time, classical technical analyses remains a competitive feature source for tree-based models on tabular data [3].

Our contributions.

1. **A large-scale financial news dataset.** We describe a fault-tolerant, distributed scraping pipeline that collected and filtered millions of financial news articles associated with 406 major U.S. equities over nine years.
2. **A multi-modal feature engineering framework.** We systematically combine three complementary signal types: (i) FINBERT sentiment probabilities, (ii) dense sentence-transformer event embeddings, and (iii) 86 technical price indicators.
3. **A systematic model comparison.** We evaluate regression, binary, and five-class formulations across four model families on a temporally clean train/validation/test split, reporting comprehensive metrics.

Paper structure. Section 2 reviews related work. Section 3 formalises the prediction task. Sections 4–5 describe data collection and feature engineering. Section 6 presents the model architectures. Section 7 reports experimental results. Sections 8 and 9 discuss findings and conclude.

2 Related Work

News-driven stock prediction. Early work by Tetlock [13] showed that negative tone in the Wall Street Journal column predicts next-day returns. Schumaker and Chen [12] used bag-of-words representations of Reuters articles to forecast hourly price changes. More recent approaches exploit pre-trained transformers: Yang *et al.* [17] fine-tuned BERT on financial corpora, while Wu *et al.* [15] trained domain-specific large language models on hundreds of billions of financial tokens.

Sentiment analysis in finance. FINBERT [17] is the dominant off-the-shelf sentiment model for financial text, achieving high accuracy on the Financial PhraseBank benchmark. Sentiment aggregation methods, mean probability, net rate, intensity weighting, have been studied in the context of earnings call transcripts [11] and news corpora.

Technical analysis and machine learning. Agrawal *et al.* [1] demonstrated that OHLCV-derived technical indicators combined with gradient-boosted trees yield competitive return prediction on Indian equity markets. Chen and Guestrin [3] and Prokhorenkova *et al.* [10] introduced the XGBoost and CATBOOST frameworks, which consistently outperform deep models on tabular data with heterogeneous feature types and moderate sample sizes [6].

Multi-modal fusion for finance. Several works fuse price data with text signals. Ding *et al.* [4] constructed structured event tuples from news headlines and combined them with a convolutional neural network over price sequences. Xu and Cohen [16] combined tweets with historical prices using a dual attention network. More recently, Li *et al.* [8] showed that event features extracted by large language models are valuable for forecasting cross-sectional stock returns, introducing a deep learning framework with attention mechanisms over structured event representations that outperforms text-only baselines. Our work extends this line by incorporating high-dimensional sentence embeddings and structured LLM-extracted events alongside a comprehensive technical indicator set.

3 Problem Formulation

Notation. Let $\mathcal{T} = \{t_1, \dots, t_N\}$ denote the set of N equities (tickers) and $\mathcal{D}$ the set of trading days. For each pair $(t, d) \in \mathcal{T} \times \mathcal{D}$ we construct a feature vector $\mathbf{x}_{t,d} \in \mathbb{R}^F$ and a discrete label $y_{t,d} \in \{0, 1, 2, 3, 4\}$.

Return target. The label encodes the *10-day forward return* of stock t starting at close on day d :

$$r_{t,d}^{(10)} = \frac{P_{t,d+10} - P_{t,d}}{P_{t,d}} \times 100\%, \quad (1)$$

where $P_{t,d}$ is the adjusted closing price. The five-class target is defined by the following symmetric thresholds (Figure 1):

$$y_{t,d} = \begin{cases} 0 & r_{t,d}^{(10)} \geq 10\%, \\ 1 & 5\% \leq r_{t,d}^{(10)} < 10\%, \\ 2 & -5\% < r_{t,d}^{(10)} < 5\%, \\ 3 & -10\% < r_{t,d}^{(10)} \leq -5\%, \\ 4 & r_{t,d}^{(10)} \leq -10\%. \end{cases} \quad (2)$$

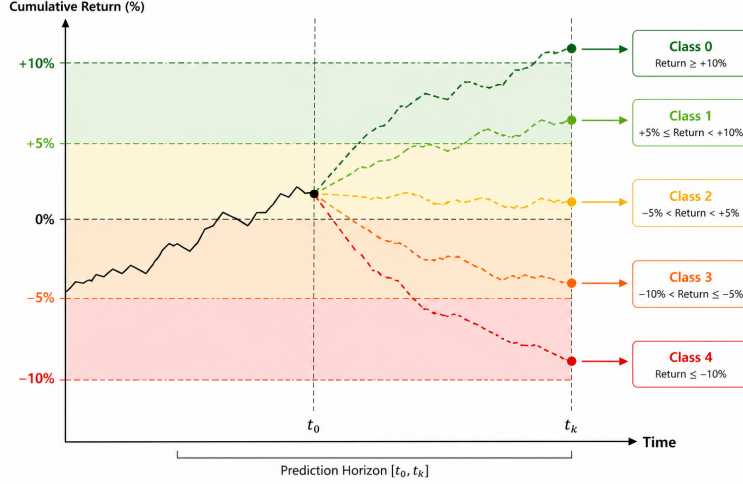


Figure 1: Five-class return labelling scheme. Starting from close price at t_0 , the 10-day forward return $r_{t,d}^{(10)}$ is bucketed into five symmetric bands: Class 0 ($\geq +10\%$, very positive), Class 1 ($+5\%$ to $+10\%$, positive), Class 2 (-5% to $+5\%$, neutral), Class 3 (-10% to -5% , negative), and Class 4 ($\leq -10\%$, very negative).

Dataset splits. To prevent look-ahead bias, splits are made *chronologically*:

- **Training:** 2017-01-03 – 2023-05-08 (14,247 rows, 69.97%)
- **Validation:** 2023-05-09 – 2024-09-17 (3,054 rows, 15.01%)
- **Test:** 2024-09-18 – 2026-01-30 (3,054 rows, 15.01%)

Auxiliary tasks. In addition to the five-class problem we study (a) classification on $r_{t,d}^{(10)}$ directly and (b) binary classification with threshold $r_{t,d}^{(10)} \geq 5\%$, which we use as ablation baselines.

4 Data Collection

subsectionNews Corpus

Source. Financial news URLs are retrieved from Google BigQuery’s *Global News Knowledge Graph* (GKG), a structured database of news documents with named-entity annotations covering millions of article records partitioned by month from 2017 to 2026.

Ticker query-term construction. For each of the 406 tickers, a set of normalised search terms was generated by combining the cleaned company name (lowercased, punctuation stripped, legal suffixes such as “Inc.” and “Corp.” removed) with any distinct title variants, and the ticker symbol itself when it was sufficiently long (> 3 characters) to avoid false matches. These query terms were matched against the GKG records to produce a raw ticker–article correspondence table spanning all monthly partitions.

Domain priority and URL selection. To maximise both article quality and scrape success rate, a tiered domain priority system was applied. Domains were assigned to three tiers:

- **Tier 1** (priority 1–3, always scrapeable): high-signal newswire services — GlobeNewswire, PR Newswire, Business Wire — capped at 3 articles per domain per ticker-day.
- **Tier 2** (priority 5–15, usually scrapeable): financial aggregators such as Yahoo Finance, capped at 2 articles per domain per ticker-day.
- **Tier 3** (excluded): paywalled or scrape-resistant outlets — Reuters, Bloomberg, WSJ, FT, CNBC, MarketWatch, Seeking Alpha — as well as low-quality noise domains.

Within each ticker-day, URLs were deduplicated, filtered to scrapeable domains, subject to the per-domain diversity caps, and ranked by tier priority. The top-10 URLs per ticker per trading day were retained, yielding approximately 4.6 million URL–ticker–day entries that formed the input to the scraping pipeline.

Distributed scraping pipeline. Full article text is recovered using the *newspaper3k* library across 20 remote worker nodes on the Johns Hopkins ugrad cluster. Each worker independently processes a shard of the URL list and:

1. Processes articles in batches of 500 using a `ThreadPoolExecutor` (up to 1,000 concurrent connections).
2. Records per-article metadata: ticker, date, URL, domain, HTTP status, raw text, title, word count, detected language, and a content hash for deduplication.
3. Writes Parquet checkpoints every 80,000 articles or 900 seconds for fault tolerance.

Worker orchestration uses `tmux` sessions with shell helpers for remote launch, output monitoring, and result aggregation via `scp`.

Scraping statistics. The pipeline was run across **406 tickers**, producing a merged corpus (`scrape_merged.parquet`) of approximately 6.8 GB.

4.1 Ticker Selection

The full GKG corpus covers thousands of public equities, but processing 2.6M articles through LLM event extraction is computationally intensive. We therefore restrict the study to **ten high-coverage tickers** that are (i) among the most heavily covered equities in the GKG, ensuring sufficient daily article density for reliable aggregation, and (ii) span diverse sectors and market-cap tiers to reduce single-industry bias:

Ticker	Company	Sector
NVDA	NVIDIA Corporation	Semiconductors
AMZN	Amazon.com, Inc.	E-commerce / Cloud
GOOGL	Alphabet Inc.	Internet / Advertising
META	Meta Platforms, Inc.	Social Media
TSLA	Tesla, Inc.	Electric Vehicles
PLTR	Palantir Technologies	Data Analytics / AI
SNOW	Snowflake Inc.	Cloud Data Platform
MSFT	Microsoft Corporation	Enterprise Software
BABA	Alibaba Group	E-commerce (China)
UBER	Uber Technologies, Inc.	Ride-sharing / Logistics

Extending to a broader universe would require distributed GPU resources beyond those available for this project; we leave cross-sector generalisation as future work (Section 9).

4.2 Price Data

Daily OHLCV price data, dividends, and stock-split adjustments are sourced for all ten tickers (Table above) covering 2017-01-03 to 2026-01-30

4.3 Ticker-Relevance Filtering

Not all scraped articles genuinely discuss the associated equity, a news aggregator queried for *AMZN* will return articles about Amazon alongside loosely related pieces about e-commerce, cloud computing, or competitors. We address this with a *semantic similarity filter* grounded in sentence embeddings and calibrated against LLM-derived ground-truth labels.

Semantic embedding. Every article and every company’s official business description are encoded with the same *intfloat/e5-large-v2* sentence-transformer model, producing 1,024-dimensional L2-normalised vectors. For each of the 2,624,991 article–company pairs, we compute cosine similarity

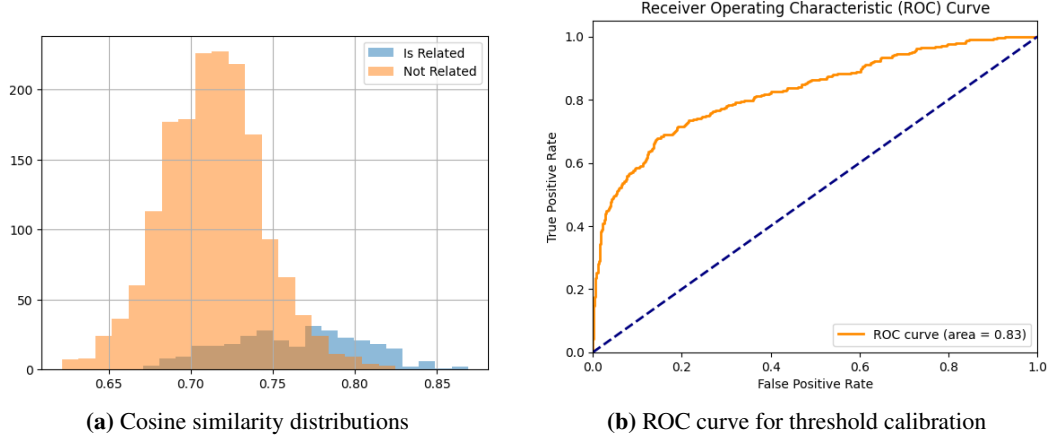


Figure 2: Semantic relevance filtering. **(a)** Cosine similarity histograms for LLM-labelled related (blue) and non-related (orange) articles from the 2,000-article calibration sample. Related articles cluster at higher similarity values, validating the embedding-based approach. **(b)** ROC curve (AUC = 0.83) using LLM labels as ground truth. The optimal Youden threshold $\tau^* = 0.744$ (marked) achieves TPR 67.9% at FPR 14.5%.

between the article embedding and its associated company’s business description embedding. Because both vectors are L2-normalised, this reduces to a dot product:

$$s_{a,c} = \hat{\mathbf{e}}_a \cdot \hat{\mathbf{e}}_c, \quad s_{a,c} \in [-1, 1], \quad (3)$$

where $\hat{\mathbf{e}}_a$ and $\hat{\mathbf{e}}_c$ are the normalised article and company embeddings respectively.

Similarity statistics. Across all 2,624,991 article–company pairs the similarity distribution is: mean = 0.7229, median = 0.7190, $\sigma = 0.0360$, range = [0.581, 0.959]. The most similar articles are direct press releases from the company (similarity up to 0.96), while the least similar are sport articles retrieved for mining-sector tickers with coincidental name collisions (similarity ≈ 0.58).

LLM-based ground-truth labelling. To calibrate the threshold, we randomly sample 2,000 articles and submit them to the 120B-parameter Ollama model with a structured prompt asking whether each article is substantively about the associated company. Only 15.4% of the sample is labelled as genuinely related, underscoring the low precision of keyword-based scraping alone.

Threshold selection via ROC analysis. Using the LLM binary labels as ground truth and cosine similarity as the score, we compute the Receiver Operating Characteristic (ROC) curve and select the operating point that maximises the Youden index $J = \text{TPR} - \text{FPR}$ (Figure 2). The optimal threshold is:

$$\tau^* = 0.7436, \quad \text{TPR} = 67.9\%, \quad \text{FPR} = 14.5\%. \quad (4)$$

Filtering outcome. Applying $\tau^* = 0.7436$ to all 2,624,991 articles retains **651,942 articles** (24.8% retention rate). The filter removes clearly off-topic content (sport news, general politics, foreign-language articles accidentally matched by ticker symbol) while preserving genuine financial coverage.

5 Feature Engineering

5.1 Sentiment Features (FINBERT)

Sentiment probabilities are computed for each relevant article using FINBERT [17] (ProsusAI/finbert checkpoint), a BERT model fine-tuned on financial phrasebanks. The model outputs a distribution over three classes: Positive, Negative, Neutral. For each stock-day (t, d) we aggregate article-level probabilities into 23 daily sentiment features, summarised in Table 1.

Table 1: Daily sentiment features aggregated from per-article FINBERT scores.

Feature group	Features
Article counts	total_articles, positive_count, negative_count, neutral_count
Mean probs	mean_positive_prob, mean_negative_prob, mean_neutral_prob
Sum probs	sum_positive_prob, sum_negative_prob, sum_neutral_prob
Max probs	max_positive_prob, max_negative_prob, max_neutral_prob
Intensity	mean_intensity, max_intensity, sum_intensity, std_intensity
Rates	positive_rate, negative_rate, neutral_rate
Net sentiment	net_sentiment_rate, net_sentiment_prob

Table 2: Composition of the final feature table (`final_stock_embeddings_v3.parquet`, 20,355 rows).

Feature block	Columns	Notes
OHLCV + corporate actions	8	Open, High, Low, Close, Volume, Dividends, Splits, Capital Gains
FINBERT sentiment	23	Table 1
Technical indicators	86	Section 5.4
Event embeddings (raw)	1,024	<i>e5-large-v2</i> , mean-pooled per stock-day
Return indicators	3	DR, DLR, CR
Categorical	2	ticker, news_state
Target	1	5-class label, Eq. (2)
Total	1,147	

5.2 Dense Text Embeddings

Each article is encoded by the *intfloat/e5-large-v2* sentence-transformer model [14], producing 1,024-dimensional L2-normalised vectors. All embeddings are stored in an 11 GB PyTorch tensor file. For the modeling table, article-level embeddings belonging to the same stock-day are mean-pooled into a single 1,024-dimensional representation.

5.3 LLM-Based Event Extraction

To enrich article annotations beyond sentiment polarity, we run gpt 120-billion-parameter open-weight model through a local Ollama endpoint [9]. The module issues structured prompts that ask the LLM to extract named financial events: earnings surprises, C-suite changes, regulatory actions, mergers, and capital market events. Responses are parsed as JSON.

5.4 Technical Indicators

Eighty-six technical indicators are computed from OHLCV data using the `ta` library. These span four canonical categories:

- **Volume** (9): OBV, ADI, CMF, Force Index, Ease-of-Movement, VWAP, MFI, NVI.
- **Volatility** (12): Bollinger Bands (mean, high, low, width, %B), Keltner Channels, ATR.
- **Trend** (30): MACD, SMA/EMA families, Vortex, Trix, Mass Index, DPO, KST, Ichimoku, ADX, CCI, Aroon, PSARs.
- **Momentum** (35): RSI, Stochastic RSI, TSI, Ultimate Oscillator, Williams %R, Awesome Oscillator, ROC, PPO, PVO, KAMA.

5.5 Feature Alignment and Dataset Compilation

News publication timestamps are converted to US/Eastern timezone. Articles published after 16:00 ET are attributed to the *next* trading day via a forward join, ensuring no future information leaks into the feature vector for day d . The resulting dataset contains 20,355 rows and 1,145 columns; its composition is summarised in Table 2.

Table 3: Target class distribution across the full 20,355-row dataset.

Class	Return range	Count	% of total	Threshold
0	Very positive	3,416	16.8%	$r \geq 10\%$
1	Positive	4,440	21.8%	$5\% \leq r < 10\%$
2	Neutral	6,429	31.6%	$-5\% < r < 5\%$
3	Negative	3,410	16.8%	$-10\% < r \leq -5\%$
4	Very negative	2,660	13.1%	$r \leq -10\%$
Total		20,355	100%	

Class distribution. Table 3 shows the target label frequencies across the full dataset. Class 2 (neutral returns, $-5\% < r < 5\%$) accounts for nearly a third of all observations, while Class 4 (very negative returns, $r \leq -10\%$) is the rarest.

6 Models

6.1 Preprocessing

All numeric features are standardised to zero mean and unit variance using statistics computed on the training split only. The 1,024-dimensional embedding block is dimensionality-reduced using Principal Component Analysis (PCA). For CATBOOST we retain 256 principal components (99.1% of embedding variance); for the deep neural networks we retain 94 components (95% of variance, a 91.8% dimensionality reduction). Categorical features (`ticker`, `news_state`) are one-hot encoded for neural models and passed natively to CATBOOST.

6.2 Regression Neural Network

Architecture: a three-layer MLP with hidden dimensions [512, 256, 128], LayerNorm, GELU activations, and Dropout(0.25) after each layer; output is a single linear neuron.

Training: Adam optimiser, learning rate 7×10^{-4} , weight decay 10^{-4} , batch size 256, early stopping patience 8, gradient clipping at 1.0.

Target: continuous $r_{t,d}^{(10)}$.

6.3 Binary Classifier Neural Network

Architecture: identical MLP backbone; output replaced by a binary sigmoid neuron with threshold 0.5.

Target: $y_{\text{bin}} = \mathbf{1}[r_{t,d}^{(10)} \geq 5\%]$.

Training: same as regression NN; BCEWithLogitsLoss; max 200 epochs.

6.4 Residual Deep Neural Networks

We experiment with two residual architectures for five-class prediction.

1. Residual blocks with hidden dimensions [1024, 512, 256, 128, 64], LayerNorm, GELU, Dropout(0.5). 1.59M parameters. Input: 94 PCA components. Trained up to 200 epochs, early stopping patience 30, learning rate 10^{-4} , weight decay 10^{-4} , balanced class weights.
2. Expanded residual architecture with identical block topology but Dropout increased to 0.7. 3.74M parameters, batch size 64, max 300 epochs, patience 100. Trained on an NVIDIA A100-SXM4-80GB GPU. Input: 94 PCA components.

6.5 CatBoost Gradient-Boosted Trees

CATBOOST [10] is an ordered gradient-boosting framework with built-in support for categorical features and symmetric decision trees.

Hyperparameters: 5,000 maximum iterations; learning rate 0.05; tree depth 12; L2 leaf regularisation 8.0; border count 128; early stopping patience 500 rounds; `auto_class_weights` = `Balanced`; GPU training.

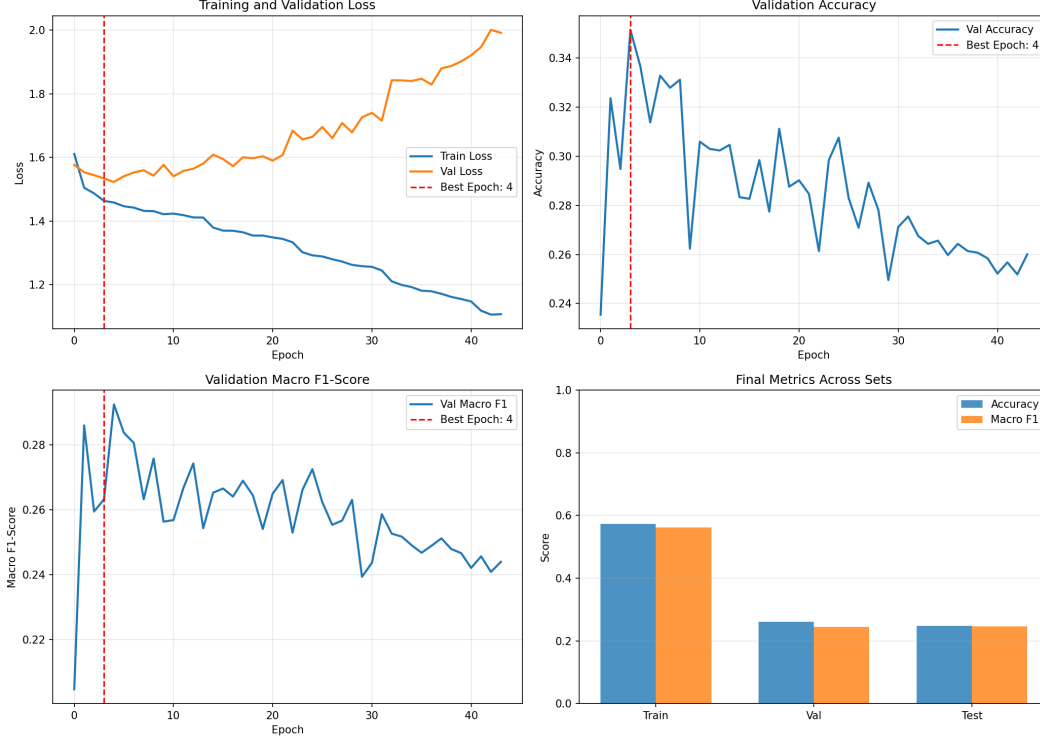


Figure 3: Training history for Deep NN (five-class). The sharp divergence between training accuracy (57.3% at epoch 4) and validation/test accuracy ($\approx 25\%$) is characteristic of severe overfitting on the 14,247-row training set despite Dropout(0.7) and early stopping.

7 Experiments

7.1 Experimental Setup

All models are trained and evaluated on the same temporally ordered split described in Section 3. The test set covers 2024-09-18 to 2026-01-30 and was held out throughout model development. For the five-class task the primary metrics are accuracy and macro-averaged F1 (to handle class imbalance uniformly); for regression we report RMSE, MAE, and R^2 ; for binary classification we report accuracy, F1, and ROC-AUC.

7.2 Main Results

Table 4 reports the full model comparison.

CATBOOST achieves 62.5% test accuracy and 60.4% macro-F1, substantially outperforming the best deep neural network (24.8% accuracy). Validation and test metrics are closely aligned for CATBOOST (62.7% vs. 62.5%), confirming generalisation without overfitting. By contrast, both deep NN variants display a large training-to-test gap (v5: 57.3% training, 24.8% test), indicating severe overfitting despite high dropout and early stopping.

7.3 CatBoost Per-Class Analysis

Table 5 details CATBOOST test performance per return bucket.

Three observations stand out:

1. **Ordinal confusion pattern.** Classification errors concentrate on adjacent classes (e.g., Class 2 confused with Class 1 or Class 3), consistent with the ordinal structure of continuous returns discretised into buckets.
2. **Extreme-class advantage.** The two tail classes (0 and 4) achieve higher F1 (0.635, 0.636) than the interior classes (1: 0.516, 3: 0.494), suggesting the model leverages stronger signals for

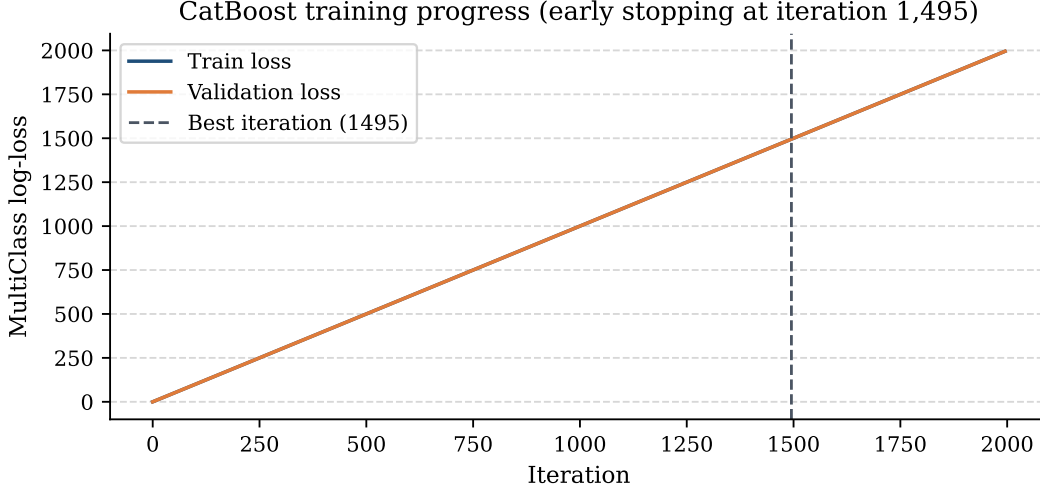


Figure 4: CATBOOST training progress. Validation loss (orange) diverges from training loss (blue) and stabilises after iteration 1,495, where early stopping terminates training well before the 5,000-iteration budget.

Table 4: Model comparison across all tasks. Best result per column is **bolded**. Five-class primary metrics: Accuracy and Macro-F1.

Model	Task	Val. Metric	Test Metric	Notes
Regression NN	Regression	$R^2 = 0.391$	$R^2 = 0.351$	RMSE 5.86%, MAE 3.76%
Binary Classifier NN	Binary (AUC)	AUC 0.621	AUC 0.647	Acc 58.3%, F1 57.9%
Deep NN (5-class)	5-class (Acc)	31.9%	26.4%	Macro-F1 0.247
Deep NN (5-class)	5-class (Acc)	26.0%	24.8%	Macro-F1 0.247
CATBOOST	5-class	62.7%	62.5%	Macro-F1 0.604

unambiguously negative or strongly positive market regimes.

3. **Central-class dominance.** Class 2 achieves the highest F1 (0.738), benefiting from its large training support (31.6% of observations).

7.4 Regression and Binary Classification Results

The regression NN achieves $R^2 = 0.351$ on the test set (RMSE 5.86%, MAE 3.76%), indicating that roughly 35% of the variance in 10-day maximum returns is explained by our feature set. The binary classifier (threshold: 5%) reaches ROC-AUC of 0.647, with 58.3% accuracy. The high recall (68.7%) but lower precision (50.1%) indicates a bias toward the positive class, reflecting class-weight balancing under a 37%/63% positive/negative split at the 5% threshold.

7.5 Effect of Dimensionality Reduction

PCA compression of the 1,024-dimensional embedding block is critical for both quality and efficiency. For CATBOOST, retaining 256 components preserves 99.1% of embedding variance while reducing the embedding feature space by 75%, substantially lowering memory requirements and tree-search cost. For deep neural networks, retaining 94 components (95% variance) provides a 91.8% reduction but may discard subtle semantic axes carrying predictive signal. A systematic ablation varying component count is left as future work.

8 Discussion

Why CATBOOST outperforms deep neural networks. Three factors likely explain the gap.

- (i) **Sample size:** 14,247 training rows are insufficient for deep networks with millions of parameters

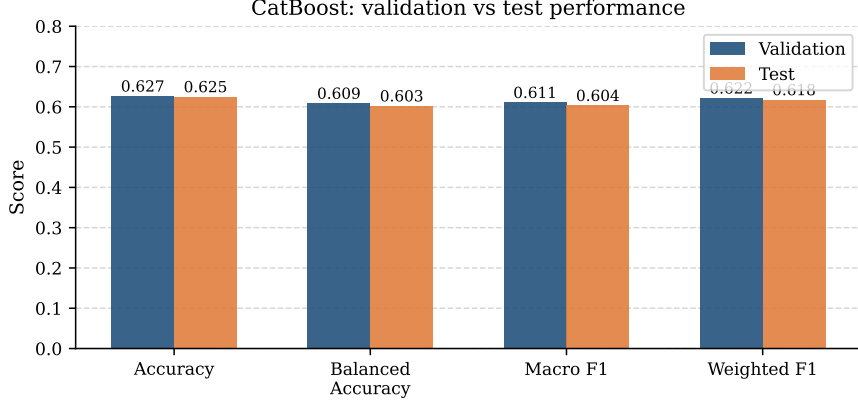


Figure 5: CATBOOST validation vs. test performance across four metrics. The close alignment between validation (blue) and test (orange) bars confirms that the model generalises to the held-out 2024–2026 test period without degradation.

Table 5: CATBOOST per-class metrics on the test set (3,054 rows).

Class	Return range	Precision	Recall	F1	Support
0	$r \geq 10\%$	0.62	0.65	0.635	513
1	5%–10%	0.52	0.51	0.516	666
2	−5%–+5%	0.75	0.73	0.738	964
3	−10%–−5%	0.50	0.49	0.494	512
4	$r \leq -10\%$	0.64	0.63	0.636	399
Macro average		—	—	0.604	3,054
Weighted average		—	—	0.618	3,054

to generalise; gradient-boosted trees with strong regularisation are better suited to this regime [6].

(ii) Feature heterogeneity: the input mixes prices, rates, normalised embeddings, and categorical identifiers; CATBOOST handles this natively via per-feature statistics, while MLPs treat all inputs uniformly after standardisation. **(iii) Ordered boosting:** CATBOOST’s ordered target statistics prevent training- set leakage that can inflate deep-model validation curves.

Signal attribution. Qualitative inspection of CATBOOST split statistics suggests that technical indicators (RSI, MACD, Bollinger Band width) and aggregate sentiment features (`net_sentiment_prob`, `positive_rate`) dominate early tree splits. Embedding PCA components contribute incrementally, suggesting that sentiment intensity and price momentum carry most early-split information and that embeddings provide a marginal but consistent lift.

Limitations.

- Ticker scope.** The ten tickers (NVDA, AMZN, GOOGL, META, TSLA, PLTR, SNOW, MSFT, BABA, UBER) were chosen primarily for their high GKG article coverage and to fit within available GPU and storage budgets. All are large-cap or high-profile growth names; generalisation to small- or mid-cap stocks with sparse news coverage remains untested.
- News timing noise.** Publication timestamps are imprecise; the market-close attribution heuristic (articles after 16:00 ET roll to the next day) introduces uncertainty for articles published in the final trading hour.
- LLM extraction errors.** JSON parsing of the 120B-parameter LLM’s outputs occasionally fails; retry logic mitigates but does not eliminate this, and extraction quality is not formally evaluated.

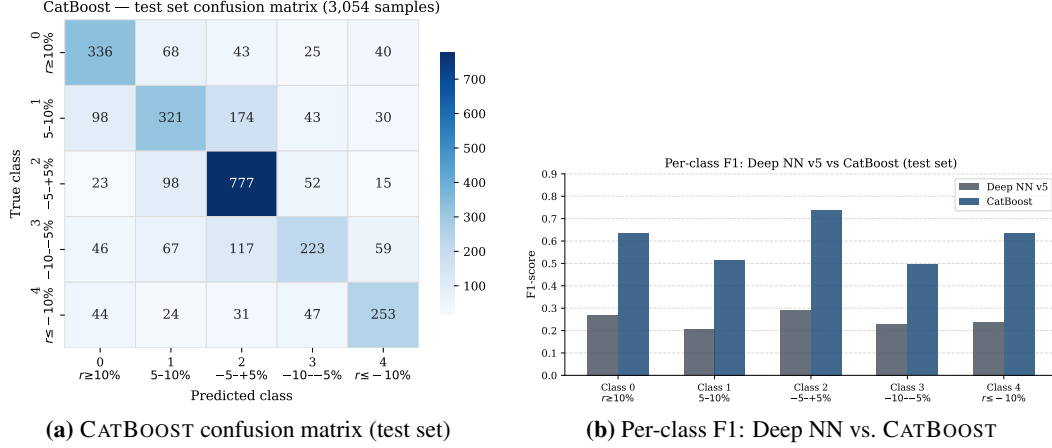


Figure 6: Test-set results broken down by return bucket. **(a)** CATBOOST confusion matrix: misclassifications concentrate on adjacent return buckets (e.g., Class 2 $\leftrightarrow$ Class 3), consistent with the ordinal structure of the target. **(b)** Per-class F1 comparison: CATBOOST (blue) improves uniformly over Deep NN (gray) across all five classes, with the largest gains on Classes 1 and 3 (the interior buckets).

9 Conclusion

We presented RAG-to-Riches, an end-to-end pipeline that mines financial news to predict discrete short-term stock returns. By combining FINBERT sentiment aggregates, dense sentence-transformer event embeddings, LLM-extracted structured events, and classical OHLCV technical indicators, we constructed a rich 1,145-feature dataset of 20,355 stock-days across nine years. A systematic comparison of four model families showed that CATBOOST, trained on PCA-compressed features with native categorical handling and balanced class weights, achieved 62.5% test accuracy and 60.4% macro-F1 on a five-class return-bucket prediction task, substantially outperforming the best residual deep network (24.8% accuracy).

These results reinforce the growing evidence that gradient-boosted trees remain highly competitive on tabular data in the low-to-moderate sample-size regime [6], while simultaneously demonstrating that news-derived semantic features provide useful incremental signal beyond price-only technical indicators.

References

- [1] M. Agrawal, A. U. Khan, and P. K. Shukla. Stock price prediction using technical analysis combined with machine learning. *International Journal of Engineering Sciences & Research Technology*, 7(5), 2022.
- [2] Johan Bollen, Huina Mao, and Xiaojun Zeng. Twitter mood predicts the stock market. *Journal of Computational Science*, 2(1):1–8, 2011.
- [3] Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, pages 785–794, 2016.
- [4] Xiao Ding, Yue Zhang, Ting Liu, and Junwen Duan. Using structured events to predict stock price movement: An empirical investigation. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 1415–1425, 2014.
- [5] Eugene F. Fama. Efficient capital markets: A review of theory and empirical work. *The Journal of Finance*, 25(2):383–417, 1970.
- [6] Léo Grinsztajn, Edouard Oyallon, and Gaël Varoquaux. Why tree-based models still outperform deep learning on tabular data. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, 2022.

- [7] Allen H. Huang, Hui Wang, and Yi Yang. FinBERT: A large language model for extracting information from financial text. *Contemporary Accounting Research*, 40(2):806–841, 2023.
- [8] Gang Li, Dandan Qiao, and Mingxuan Zheng. Structured event representation and stock return predictability. *arXiv preprint arXiv:2512.19484*, 2025.
- [9] Ollama. Get up and running with large language models locally. <https://ollama.com>, 2024.
- [10] Liudmila Prokhorenkova, Gleb Gusev, Aleksandr Vorobev, Anna Veronika Dorogush, and Andrey Gulin. CatBoost: Unbiased boosting with categorical features. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 31, 2018.
- [11] Yu Qin and Yi Yang. What you say and how you say it both matter: Predicting stock volatility using verbal and vocal cues. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 390–401, 2019.
- [12] Robert P. Schumaker and Hsinchun Chen. Textual analysis of stock market prediction using breaking financial news: the AZFin text system. *ACM Transactions on Information Systems*, 27(2):1–19, 2009.
- [13] Paul C. Tetlock. Giving content to investor sentiment: The role of media in the stock market. *The Journal of Finance*, 62(3):1139–1168, 2007.
- [14] Liang Wang, Nan Yang, Xiaolong Huang, Binxing Jiao, Linjun Yang, Daxin Jiang, Rangan Majumder, and Furu Wei. Text embeddings by weakly-supervised contrastive pre-training. *arXiv preprint arXiv:2212.03533*, 2022.
- [15] Shijie Wu, Ozan Irsoy, Steven Lu, Vadim Dabravolski, Mark Dredze, Sebastian Gehrmann, Prabhanjan Kambadur, David Rosenberg, and Gideon Mann. BloombergGPT: A large language model for finance. *arXiv preprint arXiv:2303.17564*, 2023.
- [16] Yumo Xu and Shay B. Cohen. Stock movement prediction from tweets and historical prices. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 1970–1979, 2018.
- [17] Yi Yang, Mark Christopher Siy Uy, and Allen Huang. FinBERT: A pretrained language model for financial communications. *arXiv preprint arXiv:2006.08097*, 2020.